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Type: Recorded Interview

Interviewee: Dr. Morsi Ahmed

Position: Head of the Material Science and Engineering Department, Chair of the Environmental Engineering Program, Abdullah Salem University

Interviewers : Retaj Bader , Shahed Yousef and Nada Ibrahim

[Recording Begins]

Interviewer:

Good afternoon, Dr. Morsi. It's a pleasure to meet you.

Dr. Morsi Ahmed:

Nice meeting you too.

Interviewer:

Before we start, could you please introduce yourself?

Dr. Morsi Ahmed:

Of course. I'm Dr. Morsi Ahmed, the Head of the Material Science and Engineering Department. I'm also serving as the Chair of the Environmental Engineering Program here at Abdullah Salem University. I graduated from Virginia Tech and spent 7 years there. After that, I moved to Germany, where I worked at the Karlsruhe Institute of Technology for another 7 years. Then I joined King Fahd University of Petroleum and Minerals for 7 years. Finally, in 2024, I moved here to Abdullah Salem University.

Interviewer: That's an impressive background, Dr. Morsi. Thank you. My first question is: In your opinion, what are the main challenges that countries face when trying to shift from traditional energy sources to renewable ones?

Dr. Morsi Ahmed:

That's a very good and broad question. I'll try to summarize it. There are several challenges, especially in the Gulf region. The most important challenge is infrastructure. All the infrastructure in the Gulf area is designed for traditional oil and gas. Moving from oil and gas to renewable energy requires large investments and new infrastructure, which affects the economic side. The second challenge is technical. The Gulf region faces dust and high temperatures, both of which reduce the efficiency of renewable energy systems such as solar panels. Dust accumulates on panels and decreases performance, while high temperature also lowers their efficiency. Another challenge is cost. Currently, the prices of renewable energy are still not as competitive as traditional energy sources. So, to summarize — the main challenges are: infrastructure and economic factors, technical limitations (dust and high temperatures), and cost competitiveness.

Interviewer:

That's very clear, thank you. My second question is: What kind of materials are the most promising for improving renewable energy technologies, like solar panels or wind turbines?

Dr. Morsi Ahmed:

From a materials science point of view, there are several promising materials. First, Perovskites these materials can significantly increase the efficiency of solar panels. Second, carbon-based composites these materials are lightweight and resistant to corrosion, making them ideal for wind energy systems. Finally, MOFs, or Metal-Organic Frameworks. This material was invented by a Nobel laureate, Omar Yaghi, at the University of California, Berkeley. MOFs can store hydrogen, which makes them very important for green hydrogen production and the future of renewable energy. So, these three materials Perovskites, carbon-based composites, and MOFs are strong candidates for improving renewable energy technologies.

Interviewer:

That's fascinating, Dr. Morsi. Thank you very much for your time and insightful answers.

Dr. Morsi Ahmed: You're welcome. It was my pleasure.

[Recording Ends]

Reference :

Ahmed, M. (2025, November 3). Recorded interview on renewable energy challenges and materials. Conducted by R. Bader, S. Yousef, & N. Ibrahim. Abdullah Salem University. (Personal communication, not publicly available.)

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